s02_qc_merge_genetic_dist

2025-03-03

```
res_dir <- "../results/s02_qc_merge_genetic_dist_malagasy/"
```

This script repeats the steps in s01a_qc_merge to merge malagasy data, while retaining and ordering the genetic distance of SNPs, for ALDER analysis.

Generate gen tibble

```
if (file.exists("../data/raw/data_input/Madagascar_full.gt")) {
  madagascar <- gt_load("../data/raw/data_input/Madagascar_full.gt")
} else {
  madagascar <- gen_tibble("../data/raw/data_input/Madagascar_full.bed", quiet = TRUE, valid_alleles = c
}

cat("Number of individuals:", nrow(madagascar), "\n")
```

```
## Number of individuals: 192
```

```
cat("Number of loci:", count_loci(madagascar))
```

```
## Number of loci: 713014
```

```
if (file.exists("../data/raw/data_input/Madagascar_full_v2.bk")) {
  file.remove("../data/raw/data_input/Madagascar_full_v2.gt")
  file.remove("../data/raw/data_input/Madagascar_full_v2.bk")
  file.remove("../data/raw/data_input/Madagascar_full_v2.rds")
}
```

Our raw dataset has 713014 SNPs and 192 individuals

Add metadata

```
# Read in Pierron Malagasy metadata
pop_table <- read.table("../data/raw/data_input/Madagascar_Borneo.poptable",
  sep = "\t", fill = TRUE
)
colnames(pop_table) <- c("IID", "V2", "FID", "Population", "V5", "V6", "V7")
```

```
# Add "RR" to pop_table$IID
pop_table$IID <- paste0(pop_table$IID, "RR")

# Find Malagasy entries
pop_table_sub <-
  subset(
    pop_table,
    pop_table$IID %in% madagascar$id & pop_table$V5 == "Madagascar"
  )

# match up the ID's with pop table metadata
matching_m <- match(madagascar$id, pop_table_sub$IID)

madagascar$population <-
  ifelse(!is.na(matching_m),
    pop_table_sub$Population[matching_m], madagascar$population
  )
```

Remove the random "I" loci

```
remove <- which(show_loci(madagascar)$allele_ref == "I")
madagascar <- madagascar %>% select_loci(-all_of(c(remove)))
count_loci(madagascar)
```

```
## [1] 713013
```

Subset to autosomes only

```
madagascar <-
  madagascar %>% select_loci_if(show_loci(genotypes)$chromosome %in% c(1:22))
count_loci(madagascar)
```

```
## [1] 693051
```

Find and remove duplicates

```
dups <- find_duplicated_loci(madagascar)
madagascar <- madagascar %>% select_loci(-all_of(dups))
count_loci(madagascar)
```

```
## [1] 693045
```

```
# remove entries with genetic distance == 0
no_dist <- which(show_loci(madagascar)$genetic_dist == 0)
madagascar <- madagascar %>% select_loci(-all_of(no_dist))
count_loci(madagascar)
```

```
## [1] 691064
```

Malagasy Pierron data

```
if (file.exists("../data/raw/data_input/Pierron_Malagasy.gt")) {
  madagascar_p <- gt_load("../data/raw/data_input/Pierron_Malagasy.gt")
} else {
  madagascar_p <-
    gen_tibble("../data/raw/data_input/Pierron_Malagasy.bed",
      quiet = TRUE,
      allow_duplicates = TRUE
    )
}

cat("Number of individuals:", nrow(madagascar_p), "\n")
```

```
## Number of individuals: 69
```

```
cat("Number of loci:", count_loci(madagascar_p))
```

```
## Number of loci: 661324
```

Append population info

```
pop_table <- read.table("../data/raw/data_input/Madagascar_Borneo.poptable",
  sep = "\t", fill = TRUE
)
colnames(pop_table) <- c("IID", "V2", "FID", "Population", "V5", "V6", "V7")

pop_table_sub <-
  subset(
    pop_table,
    pop_table$IID %in% madagascar_p$id & pop_table$V5 == "Madagascar"
  )

matching_m <- match(madagascar_p$id, pop_table_sub$IID)
madagascar_p$population <- ifelse(!is.na(matching_m),
  pop_table_sub$Population[matching_m],
  madagascar_p$population
)
```

Find and remove duplicates

```
dups <- find_duplicated_loci(madagascar_p)
madagascar_p <- madagascar_p %>% select_loci(-all_of(dups))

# remove entries with no allele_ref & no allele_alt
no_allele <- which(is.na(show_loci(madagascar_p)$allele_ref) & is.na(show_loci(madagascar_p)$allele_alt)
madagascar_p <- madagascar_p %>% select_loci(-all_of(no_allele))
count_loci(madagascar_p)
```

```
## [1] 658101
```

```
if (file.exists("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge.gt")) {
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge.gt")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge.bk")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge.rds")
}
```

```
## [1] TRUE
```

```
if (file.exists("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_v2.bk")) {
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_v2.gt")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_v2.bk")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_v2.rds")
}
```

```
## Warning in
## file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_v2.gt"):
## cannot remove file
## '../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_v2.gt', reason
## 'No such file or directory'
```

```
## [1] TRUE
```

Merge the two datasets

```
malagasy_merged <- rbind(madagascar, madagascar_p,
  as_is = FALSE, flip_strand = TRUE,
  use_position = FALSE, quiet = FALSE,
  backingfile = "../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 691064 reduced to 642520
```

```
## ( 1954 are ambiguous, of which 1954  were removed)
## ----------------------------
## dataset: target
## number of SNPs: 658101 reduced to 642520
## ( 171 were flipped to match the reference set)
## ( 0 are ambiguous, of which 0 were removed)

##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbi

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dis
```

```
if (file.exists("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_v2.bk")) {
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_v2.bk")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_v2.rds")
}
```

Sort the order

```
# reorder the loci
malagasy_merged <- gt_order_loci(malagasy_merged,
  use_current_table = FALSE,
  ignore_genetic_dist = FALSE
)
```

```
##
## gen_backing files updated, now

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!
```

Quality control on the merged dataset

```
malagasy_merged <- malagasy_merged %>% group_by(population)
loci_report_m <- qc_report_loci(malagasy_merged)
malagasy_merged <- malagasy_merged %>% ungroup()
```

Filter loci

```
to_keep <- which(loci_report_m$hwe_p > 0.01)
malagasy_merged <- malagasy_merged %>% select_loci(all_of(to_keep))
count_loci(malagasy_merged)
```

```
## [1] 638245
```

```
malagasy_merged <-
  malagasy_merged %>% select_loci_if(loci_missingness(genotypes) < 0.01)
count_loci(malagasy_merged)
```

```
## [1] 620093
```

```
malagasy_merged <-
  malagasy_merged %>% select_loci_if(loci_maf(genotypes) > 0.025)
count_loci(malagasy_merged)
```

```
## [1] 579780
```

Filter individuals

```
malagasy_merged <- malagasy_merged %>% group_by(population)
individual_report_m <-
  qc_report_indiv(malagasy_merged, kings_threshold = 0.0884)
malagasy_merged <- malagasy_merged %>% ungroup()
malagasy_merged <-
  malagasy_merged %>%
  filter(id %in% individual_report_m$id & individual_report_m$to_keep == TRUE)
malagasy_merged <-
  malagasy_merged %>% filter(indiv_missingness(genotypes) < 0.1)
```

Loci and sample count

```
nrow(malagasy_merged)
```

```
## [1] 233
```

```
count_loci(malagasy_merged)
```

```
## [1] 579780
```

```
if (file.exists("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean.gt")) {
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean.gt")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean.bk")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean.rds")
}
```

```
## [1] TRUE
```

```
gt_save(malagasy_merged,
  file_name = "../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean"
)
```

```
##
## gen_tibble saved to ../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean.gt

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean.gt')

## [1] "../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dist/ma
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dist/ma
```

```
malagasy_merged <-
  gt_update_backingfile(malagasy_merged,
    backingfile = "../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean"
  )
```

```
##
## gen_backing files updated, now

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!
```

Impute

```
malagasy_merged <- gt_impute_simple(malagasy_merged, method = "mean")
```

LD clumping

```
to_keep_LD <- loci_ld_clump(malagasy_merged,
  thr_r2 = 0.2, size = 100,
  return_id = TRUE, use_positions = TRUE
)

malagasy_merged_ld <- malagasy_merged %>% select_loci(all_of(to_keep_LD))
```

Loci and sample count

```
nrow(malagasy_merged_ld)
```

```
## [1] 233
```

```
count_loci(malagasy_merged_ld)
```

```
## [1] 223918
```

```
if (file.exists("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean_ld.gt")) {
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean_ld.gt")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean_ld.bk")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean_ld.rds")
}
```

```
## [1] TRUE
```

```
gt_save(malagasy_merged_ld,
  file_name = "../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean_ld"
)
```

```
##
## gen_tibble saved to ../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean_ld.gt

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:
```

```
## gt_load('../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean_ld.gt')

## [1] "../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean_ld.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dist/mal
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dist/mal
```

```
malagasy_merged_ld <-
  gt_update_backingfile(malagasy_merged_ld,
    backingfile = "../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean_ld"
  )
```

```
##
## gen_backing files updated, now

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!
```

Load Borneo data

```
if (file.exists("../data/raw/data_input/Borneo_raw_data.gt")) {
  borneo <- gt_load("../data/raw/data_input/Borneo_raw_data.gt")
} else {
  borneo <-
    gen_tibble("../data/raw/data_input/Borneo_raw_data.bed",
      valid_alleles = c("A", "T", "C", "G", "I", "N"),
      quiet = TRUE,
      allow_duplicates = TRUE
    )
}

borneo$population <- "Borneo"
```

Find and remove duplicates

```
dups <- find_duplicated_loci(borneo)
borneo <- borneo %>% select_loci(-all_of(dups))
```

```
if (file.exists("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_borneo.gt")) {
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_borneo.gt")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_borneo.bk")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_borneo.rds")
}
```

```
## [1] TRUE
```

Merge

```
malagasy_merged <-
  gt_load("../data/intermediate/rbind_data/rbind_sort_dist/malagasy_merge_clean.gt")
malagasy_borneo <- rbind(malagasy_merged, borneo,
  as_is = FALSE, flip_strand = TRUE,
  use_position = FALSE, quiet = FALSE,
  backingfile = "../data/intermediate/rbind_data/rbind_sort_dist/malagasy_borneo"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 579780 reduced to 510862
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 715316 reduced to 510862
## ( 93588 were flipped to match the reference set)
## ( 1798 are ambiguous, of which 1798 were removed)

##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbi

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dis
```

Load 1000 Genomes data

Cutting down to populations of interest

```
if (file.exists("../data/raw/data_input/1000_Genomes_b37_clean.gt")) {
  g1000 <- gt_load("../data/raw/data_input/1000_Genomes_b37_clean.gt")
} else {
  g1000 <-
    gen_tibble("../data/raw/data_input/1000_Genomes_b37_clean.bed",
      quiet = TRUE,
      allow_duplicates = TRUE
    )
}
```

```
# append population info
pop_table <- read.table("../data/raw/data_input/Madagascar_Borneo.poptable",
  sep = "\t", fill = TRUE
)
colnames(pop_table) <- c("IID", "V2", "FID", "Population", "V5", "V6", "V7")

pop_table_sub <- subset(pop_table, pop_table$IID %in% g1000$id)

matching_m <- match(g1000$id, pop_table_sub$IID)
g1000$population <- ifelse(!is.na(matching_m),
  pop_table_sub$V2[matching_m],
  g1000$population
)

pop <- c("KHV", "BEB", "YRI", "PJL", "GWD", "MSL", "ESN", "LWK", "CHS")
#- Kinh in Ho Chi Minh City, Vietnam
#- Bengali in Bangladesh
#- Han Chinese South
#- Luhya in Webuye, Kenya
#- Yoruba in Ibadan, Nigeria
#- Punjabi in Lahore, Pakistan
#- Gambian in Western Division, The Gambia - Mandinka
#- Mende in Sierra Leone
#- Esan in Nigeria

g1000 <- g1000 %>% filter(population %in% pop)

count_loci(g1000)
```

```
## [1] 670552
```

```
nrow(g1000)
```

```
## [1] 882
```

Find and remove duplicates

```
dups <- find_duplicated_loci(g1000)
g1000 <- g1000 %>% select_loci(-all_of(dups))
```

Merge

```
if (file.exists("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000.gt")) {
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000.gt")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000.bk")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000.rds")
}
```

```
## [1] TRUE
```

```
mad_bor_g1000 <- rbind(malagasy_borneo, g1000,
  as_is = FALSE, flip_strand = TRUE,
  use_position = FALSE, quiet = FALSE,
  backingfile = "../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 510862 reduced to 502601
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 670552 reduced to 502601
## ( 91213 were flipped to match the reference set)
## ( 3 are ambiguous, of which 3 were removed)

##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbi

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dis
```

Load Sumba data

```
if (file.exists("../data/raw/data_input/GSE76645_INDONESIA_SUMBA_TIMOR.gt")) {
  sumba <- gt_load("../data/raw/data_input/GSE76645_INDONESIA_SUMBA_TIMOR.gt")
} else {
  sumba <-
    gen_tibble("../data/raw/data_input/GSE76645_INDONESIA_SUMBA_TIMOR.bed",
      quiet = TRUE,
      allow_duplicates = TRUE
    )
}

sumba$population <- "Sumba"

timor <- read.table("../data/raw/data_input/timor")
timor$population <- "Timor"
```

```
matching_m <- match(sumba$id, timor$V2)
sumba$population <- ifelse(!is.na(matching_m),
  timor$population[matching_m],
  sumba$population
)

count_loci(sumba)
```

```
## [1] 713246
```

```
nrow(sumba)
```

```
## [1] 240
```

```
set.seed(1)

subsample <- function(x) {
  if (nrow(x) <= 30) {
    return(x)
  } else {
    sample_n(x, size = 30)
  }
}

sumba_only <- sumba %>%
  group_by(population) %>%
  group_modify(~ subsample(.x)) %>%
  ungroup()

sumba <- sumba %>% filter(id %in% sumba_only$id)

nrow(sumba)
```

```
## [1] 37
```

Find and remove duplicates

```
dups <- find_duplicated_loci(sumba)
sumba <- sumba %>% select_loci(-all_of(dups))
```

Merge

```
if (file.exists("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum.gt")) {
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum.gt")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum.bk")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum.rds")
}
```

```
## [1] TRUE
```

```
mad_bor_g1000_sum <- rbind(mad_bor_g1000, sumba,
  as_is = FALSE, flip_strand = TRUE,
  use_position = FALSE, quiet = FALSE,
  backingfile = "../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 502601 reduced to 502585
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 713246 reduced to 502585
## ( 91211 were flipped to match the reference set)
## ( 2056 are ambiguous, of which 2056 were removed)

##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbi

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dis
```

Load NEO data

```
if (file.exists("../data/intermediate/rbind_data/NEO_full.gt")) {
  NEO_full <- gt_load("../data/intermediate/rbind_data/NEO_full.gt")
} else {
  NEO_full <-
    gen_tibble("../data/raw/data_input/NEO_full.bed",
      quiet = TRUE,
      allow_duplicates = TRUE
    )
}
```

Find and remove duplicates

```
dups <- find_duplicated_loci(NEO_full)
NEO_full <- NEO_full %>% select_loci(-all_of(dups))
```

Merge

```
if (file.exists("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo.gt")) {
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo.gt")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo.bk")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo.rds")
}
```

```
## [1] TRUE
```

```
mad_bor_g1000_sum_neo <- rbind(mad_bor_g1000_sum, NEO_full,
  as_is = FALSE,
  flip_strand = TRUE,
  quiet = FALSE,
  use_position = TRUE,
  backingfile = "../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 502585 reduced to 418639
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 1400891 reduced to 418639
## ( 1 were flipped to match the reference set)
## ( 0 are ambiguous, of which 0 were removed)

##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbi

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dis
```

Quality control on the merged dataset

```
mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% group_by(population)
loci_report_m <- qc_report_loci(mad_bor_g1000_sum_neo)
mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% ungroup()
```

Loci

```
to_keep <- which(loci_report_m$hwe_p > 0.01)
mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% select_loci(all_of(to_keep))
count_loci(mad_bor_g1000_sum_neo)
```

```
## [1] 417933
```

```
mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% select_loci_if(loci_missingness(genotypes) < 0.01)
count_loci(mad_bor_g1000_sum_neo)
```

```
## [1] 237401
```

```
mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% select_loci_if(loci_maf(genotypes) > 0.025)
count_loci(mad_bor_g1000_sum_neo)
```

```
## [1] 235715
```

Filter indiv

```
# temporarily remove all populations with only one individual
pop_tbl <- table(mad_bor_g1000_sum_neo$population)
to_remove <- which(pop_tbl == 1)
to_remove_pops <- names(to_remove) # this removes a lot of DRC samples and some Mozambique samples (rel

removed_pops <- mad_bor_g1000_sum_neo %>%
  filter(population %in% to_remove_pops)

mad_bor_g1000_sum_neo <-
  mad_bor_g1000_sum_neo %>% filter(!population %in% to_remove_pops)


# filter indivs
mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% group_by(population)
individual_report_m <-
  qc_report_indiv(mad_bor_g1000_sum_neo, kings_threshold = 0.0884)
mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% ungroup()
mad_bor_g1000_sum_neo <-
  mad_bor_g1000_sum_neo %>% filter(id %in% individual_report_m$id & individual_report_m$to_keep == TRUE
```

```
if(file.exists("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_rebind.bk")){
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_rebind.bk")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_rebind.gt")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_rebind.rds")
}
```

```
## [1] TRUE
```

```
# bind them together again
mad_bor_g1000_sum_neo <- rbind(mad_bor_g1000_sum_neo, removed_pops, backingfile = "../data/intermediate/
```

```
## harmonising loci between two datasets
## flip_strand =  FALSE  ; remove_ambiguous =  FALSE
## ------------------------------
## dataset: reference
## number of SNPs: 235715 reduced to 235715
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 235715 reduced to 235715
## ( 0 were flipped to match the reference set)
## ( 0 are ambiguous, of which 0 were removed)

##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbi

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dis
```

```
NEO_full <-
  NEO_full %>% filter(indiv_missingness(genotypes) < 0.1)

# remove all populations with only one individual
# pop_tbl <- table(mad_bor_g1000_sum_neo$population)
# to_remove <- which(pop_tbl == 1)
# to_remove_pops <- names(to_remove) # this removes a lot of DRC samples and some Mozambique samples (r
# mad_bor_g1000_sum_neo <-
#   mad_bor_g1000_sum_neo %>% filter(!population %in% to_remove_pops)

# run filtering
# mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% group_by(population)
# individual_report_m <-
#   qc_report_indiv(mad_bor_g1000_sum_neo, kings_threshold = 0.0884)
```

```
# mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% ungroup()
# mad_bor_g1000_sum_neo <-
#   mad_bor_g1000_sum_neo %>% filter(id %in% individual_report_m$id & individual_report_m$to_keep == TR
mad_bor_g1000_sum_neo <-
  mad_bor_g1000_sum_neo %>% filter(indiv_missingness(genotypes) < 0.1)
```

Loci and sample count

```
nrow(mad_bor_g1000_sum_neo)
```

```
## [1] 2742
```

```
count_loci(mad_bor_g1000_sum_neo)
```

```
## [1] 235715
```

```
if (file.exists("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_no_ld.gt")) {
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_no_ld.gt")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_no_ld.bk")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_no_ld.rds")
}
```

```
## [1] TRUE
```

```
gt_save(
  mad_bor_g1000_sum_neo,
  "../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_no_ld"
)
```

```
##
## gen_tibble saved to ../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_no_ld.gt

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_no_ld.gt')

## [1] "../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_no_ld.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dist/ma
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dist/ma
```

```
mad_bor_g1000_sum_neo <-
  gt_update_backingfile(mad_bor_g1000_sum_neo,
    backingfile = "../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_no_ld"
  )
```

```
##
## gen_backing files updated, now

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!
```

Impute

```
mad_bor_g1000_sum_neo <- gt_impute_simple(mad_bor_g1000_sum_neo,
  method = "mean"
)
```

LD clumping

```
to_keep_LD <- loci_ld_clump(mad_bor_g1000_sum_neo,
  thr_r2 = 0.2, size = 100,
  return_id = TRUE, use_positions = TRUE
)

mad_bor_g1000_sum_neo_ld <- mad_bor_g1000_sum_neo %>% select_loci(all_of(to_keep_LD))
```

Loci and sample count

```
nrow(mad_bor_g1000_sum_neo)
```

```
## [1] 2742
```

```
count_loci(mad_bor_g1000_sum_neo)
```

```
## [1] 235715
```

```
if (file.exists("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_clean_ld.gt")) {
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_clean_ld.gt")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_clean_ld.bk")
  file.remove("../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_clean_ld.rds")
}
```

```
## [1] TRUE
```

```
gt_save(
  mad_bor_g1000_sum_neo,
  "../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_clean_ld"
)
```

```
##
## gen_tibble saved to ../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_clean_ld.g

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_clean_ld.gt')

## [1] "../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_clean_ld.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dist/ma
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_sort_dist/ma
```

```
mad_bor_g1000_sum_neo <-
  gt_update_backingfile(mad_bor_g1000_sum_neo,
    backingfile = "../data/intermediate/rbind_data/rbind_sort_dist/mad_bor_g1000_sum_neo_clean_ld"
  )
```

```
##
## gen_backing files updated, now

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/rbind_so

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!
```

Check that we have genetic distances

```
any(is.na(show_loci(mad_bor_g1000_sum_neo)$genetic_dist))
```

```
## [1] FALSE
```

Recording R environment